ESTIN

Basics of Image Processing

S4 (DS-AI) 2025-2026

Lab° 02(extra)

Type the following code:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread("red_bird.png")
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

R = img_rgb[:, :, 0]
G = img_rgb[:, :, 1]
B = img_rgb[:, :, 2]

mask_rgb = (R > 150) & (G < 100) & (B < 100)
mask_rgb = mask_rgb.astype(np.uint8) * 255
result_rgb = cv2.bitwise_and(img_rgb, img_rgb, mask=mask_rgb)

img_hsv = cv2.cvtColor(img_rgb, cv2.COLOR_RGB2HSV)

lower_red1 = np.array([0, 120, 50])
upper_red1 = np.array([10, 255, 255])
lower_red2 = np.array([170, 120, 50])
upper_red2 = np.array([180, 255, 255])

mask_hsv1 = cv2.inRange(img_hsv, lower_red1, upper_red1)
mask_hsv2 = cv2.inRange(img_hsv, lower_red2, upper_red2)
mask_hsv = cv2.bitwise_or(mask_hsv1, mask_hsv2)
result_hsv = cv2.bitwise_and(img_rgb, img_rgb, mask=mask_hsv)

dark_img_rgb = cv2.convertScaleAbs(img_rgb, alpha=0.5, beta=0)
bright_img_rgb = cv2.convertScaleAbs(img_rgb, alpha=1.5, beta=0)

R_dark, G_dark, B_dark = dark_img_rgb[:,:,0], dark_img_rgb[:,:,1], dark_img_rgb[:,:,2]
mask_rgb_dark = (R_dark > 150) & (G_dark < 100) & (B_dark < 100)
mask_rgb_dark = mask_rgb_dark.astype(np.uint8) * 255
result_rgb_dark = cv2.bitwise_and(dark_img_rgb, dark_img_rgb, mask=mask_rgb_dark)
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R_bright, G_bright, B_bright = bright_img_rgb[:,:,0], bright_img_rgb[:,:,1], bright_img_rgb[:,:,2]
mask_rgb_bright = (R_bright > 150) & (G_bright < 100) & (B_bright < 100)
mask_rgb_bright = mask_rgb_bright.astype(np.uint8) * 255
result_rgb_bright = cv2.bitwise_and(bright_img_rgb, bright_img_rgb, mask=mask_rgb_bright)

dark_hsv = cv2.cvtColor(dark_img_rgb, cv2.COLOR_RGB2HSV)
mask_hsv1_dark = cv2.inRange(dark_hsv, lower_red1, upper_red1)
mask_hsv2_dark = cv2.inRange(dark_hsv, lower_red2, upper_red2)
mask_hsv_dark = cv2.bitwise_or(mask_hsv1_dark, mask_hsv2_dark)
result_hsv_dark = cv2.bitwise_and(dark_img_rgb, dark_img_rgb, mask=mask_hsv_dark)

bright_hsv = cv2.cvtColor(bright_img_rgb, cv2.COLOR_RGB2HSV)
mask_hsv1_bright = cv2.inRange(bright_hsv, lower_red1, upper_red1)
mask_hsv2_bright = cv2.inRange(bright_hsv, lower_red2, upper_red2)
mask_hsv_bright = cv2.bitwise_or(mask_hsv1_bright, mask_hsv2_bright)
result_hsv_bright = cv2.bitwise_and(bright_img_rgb, bright_img_rgb, mask=mask_hsv_bright)


fig, axes = plt.subplots(3, 5, figsize=(15,12))

# Original
axes[0,0].imshow(img_rgb); axes[0,0].set_title("Original RGB"); axes[0,0].axis('off')
axes[0,1].imshow(mask_rgb,cmap='gray'); axes[0,1].set_title("mask_rgb"); axes[0,1].axis('off')
axes[0,2].imshow(result_rgb); axes[0,2].set_title("Red Detection - RGB"); axes[0,2].axis('off')
axes[0,3].imshow(mask_hsv, cmap='gray');axes[0,3].set_title("mask_hsv"); axes[0,3].axis('off')
axes[0,4].imshow(result_hsv); axes[0,4].set_title("Red Detection - HSV"); axes[0,4].axis('off')

# Dark
axes[1,0].imshow(dark_img_rgb); axes[1,0].set_title("Dark RGB (50%)"); axes[1,0].axis('off')
axes[1,1].imshow(mask_rgb_dark,cmap='gray'); axes[1,1].set_title("mask_rgb_dark"); axes[1,1].axis('off')
axes[1,2].imshow(result_rgb_dark); axes[1,2].set_title("RGB Detection Dark"); axes[1,2].axis('off')
axes[1,3].imshow(mask_hsv_dark, cmap='gray');axes[1,3].set_title("mask_hsv_dark"); axes[1,3].axis('off')
axes[1,4].imshow(result_hsv_dark); axes[1,4].set_title("HSV Detection Dark"); axes[1,4].axis('off')

# Bright
axes[2,0].imshow(bright_img_rgb); axes[2,0].set_title("Bright RGB (150%)"); axes[2,0].axis('off')
axes[2,1].imshow(mask_rgb_bright, cmap='gray'); axes[2,1].set_title("mask_rgb_bright");
axes[2,1].axis('off')
axes[2,2].imshow(result_rgb_bright); axes[2,2].set_title("RGB Detection Bright"); axes[2,2].axis('off')
axes[2,3].imshow(mask_hsv_bright, cmap='gray'); axes[2,3].set_title("mask_hsv_bright");
axes[2,3].axis('off')
axes[2,4].imshow(result_hsv_bright); axes[2,4].set_title("HSV Detection Bright"); axes[2,4].axis('off')

plt.tight_layout()
plt.show()
```

Questions

1. Which color space detects the color more accurately?
2. What happens when image brightness increases?
3. Why does HSV perform better?
4. Which channel in HSV represents the color?